# Political Disinformation, Financial Fragility, and Growth Risk

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**Abstract**

This paper examines whether political disinformation is associated with growth and whether banking stress amplifies that association. A frozen panel of 175 countries over 2000–2025 combines expert-coded Digital Society Project indicators with macroeconomic outcomes. The index describes the political information environment rather than false financial messages. In the baseline three-year regression, 1,539 observations from 121 countries yield a cumulative-growth slope of −1.52 percentage points per index standard deviation (standard error 0.73), evaluated at mean conditioning states. The banking-stress interaction is −0.94 (standard error 0.29). Sample and conditioning choices affect these estimates. Country-block bootstrap contrasts do not distinguish the lower-tail slope from the median or upper tail. Adding the index worsens tenth-quantile forecast loss by 3.22 percent in a chronological evaluation; the gain interval includes zero. An illustrative financing model explains how signal contamination and financial fragility can interact, without implying unequal quantile slopes. The results separate conditional associations, distributional differences and incremental predictive value. They do not identify causal financial-message harm, crisis thresholds or returns to information policy.

**JEL codes:** C23, C53, D83, E32, E44, G01.
**Keywords:** disinformation, information pollution, narrative economics, growth-at-risk, financial fragility, local projections.

### Statements and Declarations

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# 1 Introduction

Misleading information can affect financial decisions through attention, beliefs and confidence. Evidence on manipulated financial news and social media during bank runs establishes the relevance of information transmission for financial markets (Clarke et al. 2021; Cookson et al. 2026; Kogan et al. 2023). Social-media amplification is not itself evidence that the amplified claims were false. A distinct macroeconomic question is whether an information environment characterized by frequent disinformation is associated with worse growth outcomes, particularly when financial conditions are fragile.

This paper investigates that question with a country panel and an illustrative financing mechanism. Its empirical contribution is a transparent assessment of the information content of a broad disinformation proxy, including conditional mean associations, quantile slopes, fragility interactions and chronological forecast evaluation. It does not estimate the causal macroeconomic effect of financial misinformation or the returns to a particular content intervention.

The mechanism combines signal contamination with an explicitly assumed noise-sensitive lending rule. The resulting investment shortfall has increasing differences in contamination and fragility: deterioration in information can matter more when financing constraints are close to binding. A separate feedback example permits multiple steady states. A scalar persistence calculation gives a conditional comparison of equal proportional changes in model parameters. These results clarify hypotheses and identify assumptions needed for policy analysis; they are not substitutes for measurement or identification.

The empirical proxy combines five DSP indicators of government and party disinformation in 175 countries over 2000–2025. It measures relative intensity in the political information environment, rather than the share of financial messages that are false. The analysis retains that distinction throughout. The local projections estimate associations with cumulative outcomes, while separate quantile regressions assess whether slopes differ across the conditional growth distribution. Financial stress is a conditioning state, not an externally assigned treatment.

The three-year cumulative-growth coefficient is $-1.52$ percentage points per standard deviation of the index at mean conditioning states. The corresponding banking-stress interaction is $-0.94$ points. Two-year conditional growth slopes are $-0.71$, $-0.55$ and $-0.82$ points at the tenth, fiftieth and ninetieth percentiles. These numbers must be evaluated with their uncertainty, sample variation and reverse-timing diagnostics, rather than interpreted as treatment effects. Joint bootstrap comparisons do not distinguish the lower-tail slope from the median or upper tail, so the data do not support a disproportionately large downside association.

The forecasting extension compares the same macro-financial benchmark with and without the index. It fits component scaling and ranks solely within each training window and respects a one-year predictor-availability lag. Across held-out target years 2015–2024, the change in tenth-quantile pinball loss is reported as a $-3.22$ percent gain, where negative values mean worse performance. The evaluated design does not support improved downside-risk prediction from adding the index. Latest-vintage source data and a short evaluation period remain material limitations.

**Related literature.** Models of misinformation and narrative formation study platform incentives, social learning and competing explanations (Acemoglu et al. 2024; Eliaz and Spiegler 2020; Mueller-Frank 2024). Experimental work documents motivated evaluation and incomplete updating after retractions (Gonçalves et al. 2026; Thaler 2024). Evidence on political misperceptions and institutional confidence reinforces the need to distinguish a political-information proxy from financial-content exposure (Cariolle et al. 2024; Guriev et al. 2021; Zhuravskaya et al. 2020). Assessments of misinformation also warn against extrapolating from vivid incidents to aggregate harm (Altay et al. 2023; Budak et al. 2024).

The macroeconomic link concerns the formation and coordination of expectations. Narrative economics emphasizes the diffusion of economically consequential stories (Shiller 2017, 2019). In a complementary formal approach, Angeletos and La'O (2013) show that imperfect communication can support fluctuations in expectations and activity even without aggregate changes in preferences or technology. These mechanisms explain why information deserves a place in macroeconomic models; they do not imply that every belief-driven fluctuation reflects false information. The present financing model isolates a narrower channel through signal noise and borrowing constraints.

Empirical work on news content also separates information from sentiment. García (2013) finds that the stock-return predictability of newspaper sentiment is concentrated in recessions. Bybee et al. (2024) show that topics extracted from business news contain information about economic activity beyond standard numerical indicators. Neither sentiment nor news attention is a measure of factual accuracy. These studies motivate state-dependent analysis and predictive evaluation, while the financial-news manipulation evidence cited above motivates the distinct question of deceptive content. The DSP proxy permits broad country coverage at the cost of a less direct connection to financial decisions.

Uncertainty and financial frictions offer a further connection. Bloom (2009) models how greater uncertainty can temporarily delay investment and hiring, while Baker et al. (2016) construct a newspaper-based measure of economic policy uncertainty. Such uncertainty can arise from real policy or economic developments without disinformation. Collateral-based amplification (Bernanke et al. 1999; Kiyotaki and Moore 1997) and endogenous learning in uncertainty traps (Fajgelbaum et al. 2017) help motivate the paper's constraint and feedback mechanisms. Here increasing differences follow from an explicit lending assumption and the distribution of financing slack; the uncertainty literature does not identify that assumption's parameters or validate the proxy mapping.

The distributional analysis builds on Adrian et al. (2019), who link financial conditions to the conditional distribution of GDP growth. Busetti et al. (2021) use expectile regressions to study Italian growth risks and decompose expected shortfall. The distinction between describing conditional risk and forecasting it is central: Brownlees and Souza (2021) find that standard volatility models are competitive with quantile-regression growth-at-risk forecasts across 24 OECD countries. This motivates benchmarking the index's incremental predictive content rather than treating a significant panel association as evidence of forecasting value. The empirical tools are local projections (Jordà 2005) and two-step panel quantile estimation (Canay 2011). The contribution is their use to distinguish fragility interactions, distributional slope differences and held-out prediction for a broad disinformation proxy, with the financing mechanism making the limits of structural interpretation explicit.

Section 2 develops the illustrative mechanism; Section 3 explains measurement; Section 4 presents estimation and validation. Sections 5 and 6 delimit risk and policy interpretations, and Section 7 concludes. The appendix provides proofs, robustness results and a synthesis of the analytical and empirical findings.

## 2 An Illustrative Mechanism and Its Limits

The model separates an observed proxy for the information environment from financial-message contamination. Its role is to organize hypotheses about conditional associations. The country panel does not identify its structural parameters or a causal response to information policy.

### 2.1 A bounded information-environment proxy

Let $M_t \in (0,1)$ denote a normalized information-environment index and $x_t = \text{logit}(M_t)$. Consider

$$x_{t+1} = c + \rho x_t + \delta s_t + u_{t+1}. \tag{1}$$

Here $c$ is an intercept in logit units, $\rho$ is conditional persistence, and $s_t$ summarizes other countries' index values. An intercept can be negative; it is not a count or physical inflow of false messages. The observed index is a rank, so neither $M_t$ nor $x_t$ measures the share of financial claims that are false. The process offers a parsimonious description of persistence in an information environment, motivated by narrative economics (Shiller 2017, 2019).

**Proposition 1** (Scalar dynamics conditional on external inputs)**.** *With $s_t = 0$, $u_t = 0$ and $0 < \rho < 1$, the unique steady state is $\mu = c/(1-\rho)$, and $x_t = \mu + \rho^t(x_0 - \mu)$. Consequently $M_t = \sigma(x_t) \in (0,1)$ and converges to $\sigma(\mu)$. With independent mean-zero innovations of variance $\sigma_u^2$, the stationary latent variance is $\sigma_u^2/(1-\rho^2)$. A deviation in the latent state has half-life $h_{1/2} = \ln(2)/[-\ln(\rho)]$, exceeding one observation period precisely when $\rho > 1/2$.*

This half-life concerns a deviation of the aggregate latent index, not the survival time of a particular narrative. In the stochastic model, $\mathbb{E}[\sigma(x)]$ generally differs from $\sigma(\mathbb{E}[x])$. The nonlinear index also need not inherit the latent state's exact half-life. For jointly evolving countries with a fixed row-stochastic regional matrix $W$, the corresponding linear system is $x_{t+1} = c + (\rho I + \delta W)x_t + u_{t+1}$. Its stability requires the spectral radius of $\rho I + \delta W$ to be below one. For nonnegative $\rho, \delta$, that radius equals $\rho + \delta$. Scalar stability alone therefore does not establish stability of an endogenous regional system.

The distinction between latent and observed persistence can be made explicit. For a deterministic initial deviation $d_0 = x_0 - \mu$, the observed deviation is

$$M_t - \sigma(\mu) = \sigma(\mu + \rho^t d_0) - \sigma(\mu). \tag{2}$$

Because $0 < \sigma'(x) = \sigma(x)\{1 - \sigma(x)\} \leq 1/4$, its absolute value is at most $\rho^t|d_0|/4$. Close to the steady state it is approximately $\sigma'(\mu)\rho^t d_0$. Thus a common latent persistence can coexist with different movements in the bounded index: the same latent deviation produces a smaller index change near either endpoint than near its midpoint. Ranking countries before taking logits further makes this a statement about relative information environments, rather than absolute quantities of false content.

Persistence also makes long-run summaries sensitive to estimation error. Writing $h(\rho) = \log(2)/[-\log(\rho)]$ gives

$$h'(\rho) = \frac{\log(2)}{\rho\{\log(\rho)\}^2} > 0, \qquad h(\rho) \sim \frac{\log(2)}{1-\rho} \quad \text{as } \rho \uparrow 1. \tag{3}$$

The half-life therefore becomes highly sensitive near unity. This explains why the baseline persistence 0.864 implies 4.7 years whereas the approximation 0.985 implies a much longer horizon. Transforming a point estimate is useful for interpretation, but it does not remove uncertainty in the estimated dynamics. An interval reaching unity cannot be mapped into a finite upper bound on a stationary half-life.

### 2.2 Contamination and financing constraints

Let $p \in [0,1]$ be the unobserved probability that a financial signal is contaminated. A signal about collateral value $V$ is $s = V + b\eta d + \varepsilon$, where $b \sim \text{Bernoulli}(p)$, $\eta \in \{-1, 1\}$ has equal probabilities, $d > 0$, and $\varepsilon$ has mean zero and variance $\sigma_\varepsilon^2$. Assume the disturbances and $V$ are mutually independent, conditional on $p$.

**Lemma 1** (Signal contamination)**.** $\mathbb{E}[s \mid V, p] = V$ *and* $\text{Var}(s \mid V, p) = \sigma_\varepsilon^2 + pd^2$.

The lemma is about signal noise conditional on the fundamental. It is not a formula for posterior uncertainty $\text{Var}(V \mid s, p)$, which also depends on the prior and the signal likelihood. To obtain a transparent financing mechanism, assume directly that lenders apply a haircut increasing in noise variance:

$$L(p) = \lambda\{\bar{V} - \gamma(\sigma_\varepsilon^2 + pd^2)\}, \qquad \lambda, \gamma \geq 0. \tag{4}$$

Restrict attention to $L(p) \geq 0$; this is a reduced-form credit rule rather than a derived Bayesian valuation. An entrepreneur with net worth $W_0$ and desired investment $I^*$ invests $\min\{I^*, W_0 + L(p)\}$. The resulting shortfall can be written

$$H(p, F) = \max\{ap + bF - \kappa_0, 0\}, \qquad a, b \geq 0, \tag{5}$$

where $a = \lambda\gamma d^2$ and $F$ summarizes financing pressure. This illustration connects noisy information and collateral constraints (Bernanke et al. 1999; Bond et al. 2012; Kiyotaki and Moore 1997).

**Proposition 2** (Increasing differences)**.** *For $p_1 \leq p_2$ and $F_1 \leq F_2$, $H(p_2, F_1) - H(p_1, F_1) \leq H(p_2, F_2) - H(p_1, F_2)$. If $ap_2 + bF - \kappa_0 \leq 0$, the incremental shortfall is zero.*

Linking this mechanism to the data requires an additional hypothesis that financial contamination $p$ increases with the broad index $M$, conditional on other determinants. This mapping is neither observed nor estimated. Increasing differences survive a monotone mapping $p(M)$, but the magnitude and a linear interaction specification do not follow from ranks alone. Moreover, Proposition 2 does not establish small mean effects or an ordering of growth-quantile slopes without assumptions about how fragility, constraints and other shocks populate the outcome distribution. Those patterns are empirical questions, not consequences of the proposition by themselves.

### 2.3 Aggregation and the meaning of a stress interaction

Heterogeneity in financing slack connects the individual constraint to a smooth aggregate relationship. Let $K$ be an entrepreneur's slack threshold, with distribution function $C_K$ and continuous density $f_K$. Hold this distribution fixed as $p$ and $F$ vary, assume $\mathbb{E}|K| < \infty$, and let $a, b$ be common nonnegative coefficients. Average shortfall is

$$\overline{H}(p, F) = \mathbb{E}_K[(ap + bF - K)_+], \qquad (z)_+ = \max\{z, 0\}. \tag{6}$$

The fixed-distribution assumption isolates the constraint mechanism; selection into borrowing or changing firm composition can create additional effects that are not represented here.

**Proposition 3** (Aggregate constraint exposure)**.** *Under these assumptions,*

$$\overline{H}_p = aC_K(ap + bF), \qquad \overline{H}_{pF} = abf_K(ap + bF) \geq 0. \tag{7}$$

*If growth obeys $y = y^0 - \chi\overline{H}(p, F) + \epsilon$ with $\chi \geq 0$, and $p = p(m)$ is differentiable and nondecreasing in an index coordinate $m$, then, holding $y^0$ and the distribution of $\epsilon$ fixed,*

$$\frac{\partial y}{\partial m} = -\chi ap'(m)C_K(ap(m) + bF), \qquad \frac{\partial^2 y}{\partial m\, \partial F} = -\chi abp'(m)f_K(ap(m) + bF) \leq 0. \tag{8}$$

The level slope depends on the fraction already constrained. The interaction depends on the density close to the constraint boundary: a rise in fragility moves additional entrepreneurs into the region where information noise restricts investment. If almost nobody is constrained, the level association can be small. If almost everybody is constrained, shortfall can be large but its interaction with further stress can be small. Strong amplification is therefore most plausible where an appreciable mass lies near the boundary.

A local expansion around reference values $(m_0, F_0)$ contains $y_m(m - m_0)$ and $y_{mF}(m - m_0)(F - F_0)$, alongside linear stress and own-variable curvature terms. The empirical interaction is a tractable approximation to this surface. Its magnitude combines the prevalence and density of constraints, the credit sensitivity $a$, growth conversion $\chi$, the unobserved mapping $p'(m)$ and the units of both regressors. The panel's linear projection also averages over heterogeneous observations and need not equal a structural derivative at a particular point. Consequently the negative interaction $-0.94$ is consistent with the mechanism's sign restriction but cannot separately estimate any of these structural objects.

**Why amplification need not imply a steeper lower tail.** Consider the additional location-scale specification

$$y = \ell(m, F) + s(m, F)\epsilon, \qquad s(m, F) > 0, \qquad Q_\tau(y \mid m, F) = \ell(m, F) + s(m, F)q_\tau, \tag{9}$$

where $\epsilon$ has a fixed continuous distribution independent of $(m, F)$ and $q_\tau$ is its $\tau$ quantile. A financing shortfall may enter $\ell$ through $-\chi\overline{H}$ without changing $s$. In that case the index shifts every conditional quantile by the same amount even if $\ell_{mF} < 0$. More generally, the index-slope difference between two quantiles is $s_m(q_\tau - q_\nu)$. With $\tau < \nu$, a more negative lower-tail slope follows from $s_m > 0$, an extra dispersion assumption that increasing differences alone does not supply. The bootstrap contrasts test whether slopes differ in the fitted quantile specification.

Their failure to distinguish the tails from the median can coexist with a negative mean-growth interaction with banking stress. This location-scale example clarifies the distinction without imposing it on the estimated regressions.

### 2.4 Feedback as a theoretical possibility

Suppose a bounded, increasing function $g : \mathbb{R} \to \mathbb{R}$ describes stress feedback in latent-state units. Consider

$$\psi(x) = c + \rho x + \varphi g(x), \qquad \varphi \geq 0. \tag{10}$$

Writing $g = G \circ \sigma$ is permissible, but the Lipschitz constant below is that of the complete composite function on the latent scale.

**Proposition 4** (Uniqueness and possible multiplicity)**.** *If $g$ is globally Lipschitz with constant $L_g$ and $0 \leq \rho + \varphi L_g < 1$, $\psi$ has a unique globally attracting fixed point. Multiplicity is possible outside that sufficient condition: with $c = -3/4$, $\rho = 1/2$, $\varphi = 2$ and $g(x) = \min\{\max\{x, 0\}, 1\}$, the fixed points are $-3/2$, $1/2$ and $5/2$, with stable outer points and an unstable middle point.*

Failure of the contraction inequality does not prove a bifurcation. A smooth fold additionally requires a fixed point satisfying $\psi'(x) = 1$, with suitable curvature and transversality conditions. The panel identifies neither $\varphi$ nor $g$, so the example establishes possibility without locating countries relative to an escalation threshold.

### 2.5 A conditional half-life comparison

**Proposition 5** (Comparative statics of the scalar latent steady state)**.** *For $c > 0$ and $0 < \rho < 1$, the elasticities of $\mu = c/(1-\rho) > 0$ are $\varepsilon_c = 1$ and $\varepsilon_\rho = \rho/(1-\rho)$. Equal infinitesimal proportional reductions in $\rho$ and $c$ reduce $\mu$ by more under a reduction in persistence precisely when $\rho > 1/2$. For $c \geq 0$, $\mu$ is convex in $\rho < 1$.*

The common derivative of the logistic transformation cancels when comparing the two marginal reductions in the deterministic index $\sigma(\mu)$ at the same baseline. This does not imply the same ranking for stochastic welfare or a multi-channel burden score. For $c = 0$ the proportional elasticity of $\mu$ is undefined. For $c < 0$, reducing $\rho$ raises $\mu$ toward zero and raises $M$ toward $1/2$; a universal pollution-reduction claim is false. The observation period must also be fixed: an annual coefficient cannot be used to rank interventions acting on daily narrative lifetimes.

Let $\eta_\rho = -d \log \rho / dB_\rho$ and $\eta_c = -d \log c / dB_c$ denote proportional parameter changes per unit of expenditure, evaluated at positive $c, \rho$. Persistence intervention has the larger marginal effect on the latent steady state only if

$$\eta_\rho \frac{\rho}{1-\rho} > \eta_c. \tag{11}$$

Equal proportional parameter changes are thus an equal-cost comparison only when $\eta_\rho = \eta_c$. Neither effectiveness schedule is identified by the country panel. Corrections, friction and authentication can affect more than one parameter, further limiting a one-to-one policy interpretation.

# 3 Data and Measurement

The starting panel is the repository's frozen merge of the Digital Society Project (DSP) v8, World Development Indicators (WDI), and Worldwide Governance Indicators (WGI). It contains 4,539 country-years for 175 countries in 2000–2025. DSP v8 itself covers 179 countries over those years (Mechkova et al. 2026); this coverage does not imply that every regression contains 175 countries. Observed outcomes and horizon-specific sample sizes are reported below and in the generated tables.

## 3.1 What the index measures

DSP combines expert assessments through a measurement model (Pemstein et al. 2018). The five components concern false-information dissemination by governments domestically and abroad, political parties domestically and abroad, and foreign governments into the country. Each component is reversed so that higher values indicate more disinformation, then standardized. Weights are respectively $(1/4, 1/8, 1/4, 1/8, 1/4)$ in that ordering. The weighted composite is transformed into its pooled percentile, using $(\text{rank} - 1/2)/n$ with average ranks for ties. The resulting $M_{r,t}$ lies strictly between zero and one.

These are indicators of the broad political information environment. They are not financial-content counts, measured exposure, belief adoption or financial-message accuracy. A value of 0.89 means approximately the 89th percentile of the reference distribution; it does not mean that 89 percent of financial messages are false. No financial-specific validation sample is available in this replication package. Domestic-only or equal weights probe index construction but cannot supply that missing construct validity. The haircut mechanism in Section 2 therefore motivates a hypothesis rather than a measured channel.

The retrospective regressions retain the pooled transformation for comparability. Raw-composite and alternative-weight estimates are reported, rather than assuming that slopes are invariant to monotone transformations. For forecasting, component means, standard deviations and the empirical CDF are re-estimated using each training window only. Evaluation observations outside the training range are clipped to its finite endpoint ranks. The DSP component data required for this procedure are included in the replication data, with source and checksum records.

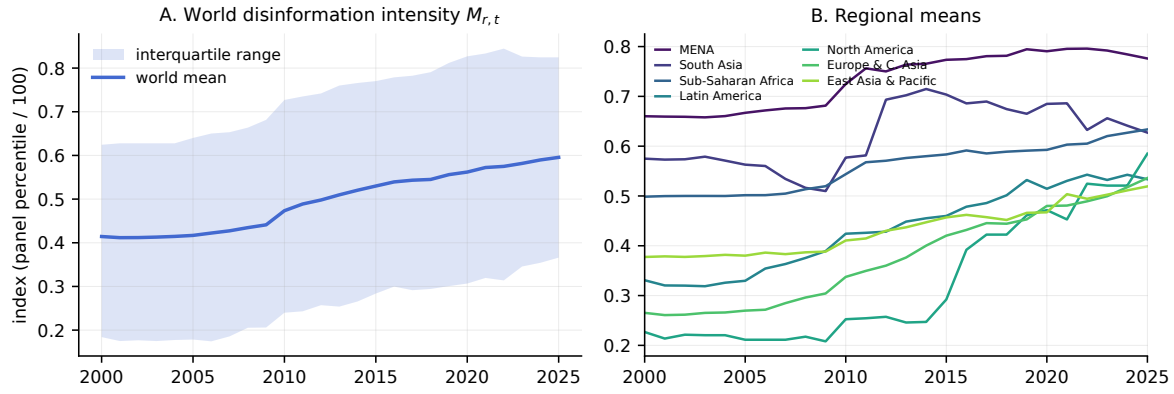



**Figure 1:** The retrospective disinformation index. The world mean, interquartile range and regional means describe relative positions in a pooled distribution, not estimated shares of false financial messages.

### 3.2 Outcomes and conditioning variables

Real GDP, gross fixed capital formation and export growth are WDI annual percentage changes (World Bank 2026). Capital formation is labeled *investment*; it does not specifically identify innovation finance. Gini and nonperforming-loan (NPL) ratios provide distributional and banking outcomes. These outcome series are never interpolated or filled in the estimation. Growth outcomes sum observed annual rates from $t$ to $t+h$; this is a cumulative percentage-point sum, not compounded GDP-level growth. Gini and NPL outcomes are observed endpoint changes from $t-1$ to $t+h$.

Conditioning states are NPLs, trade openness and inequality. Missing NPL conditioning values may use the previous observation for at most two years; Gini conditioning values may use it for at most five years. No future observation is used to fill these states, and a no-fill specification is reported separately. The frozen input includes two-sided interpolated columns for provenance; they are replaced in memory before estimation. Calendar-year matching prevents a gap from being mistaken for a one-year lag.

Controls include lagged GDP growth, log real GDP per capita, CPI inflation clipped to $[-10, 60]$, and government effectiveness. Investment and export equations also include the lagged observed outcome; lagged NPL and Gini levels already enter as conditioning states. The frozen WGI vintage ends in 2022 and its existing forward-carried values are retained through 2025; an end-2022 sample check is reported. Historical source releases and contemporaneous publication dates are not reconstructed, so the forecasting exercise uses latest-vintage data and cannot be described as real-time validation.

The data, estimation code and manuscript source are available in the public repository, together with source provenance and reproduction instructions.[1]

**Table 1:** Observed data coverage

| Variable | Mean | SD | $n$ |
|---|---|---|---|
| Disinformation index | 0.50 | 0.29 | 4,539 |
| GDP growth | 3.65 | 5.56 | 4,451 |
| Investment growth | 5.97 | 42.96 | 3,626 |
| Gini | 36.78 | 8.12 | 1,843 |
| Export growth | 6.01 | 25.42 | 3,800 |
| NPL ratio | 6.17 | 7.05 | 2,175 |

Unfilled outcome observations. Regression samples are smaller because covariates, lags and leads must be observed.

## 4 Conditional Estimates and Predictive Validation

### 4.1 Estimation and interpretation

The latent-state regression is

$$x_{r,t+1} = \alpha_r + \lambda_t + \rho x_{r,t} + \delta s_{r,t} + \theta_1 F_{r,t} + \theta_2 F_{r,t} x_{r,t} + \theta_3 \pi_{r,t} + u_{r,t+1}, \tag{12}$$

[1] https://github.com/jameskrice7/Financial-Disinformation-as-a-Macroeconomic-Risk.

where $s_{r,t}$ is the leave-one-out regional mean of the latent index. Banking stress is standardized within country in this equation. Regional co-movement is not necessarily transmission: regional economic or political shocks can affect both the index and neighboring countries.

For outcome $k$ and horizon $h = 0, \ldots, 4$, the local projection is

$$Y_{r,t,h}^{(k)} = \alpha_r^{(k,h)} + \lambda_t^{(k,h)} + \beta_h^{(k)} M_{r,t}^z + \phi_h^{(k)\top}(M_{r,t}^z W_{r,t}) + \eta_h^{(k)\top} W_{r,t} + \Gamma_h^{(k)\top} Z_{r,t}^{(k)} + e_{r,t,h}^{(k)}. \quad (13)$$

$M^z$ is the standardized index and $W$ contains standardized NPLs, trade openness and Gini measured at $t-1$. $Z$ contains clipped inflation, log GDP per capita, government effectiveness, lagged growth and, for investment and exports, the lagged observed outcome. Country and year effects are estimated by iterated within transformation. Standard errors cluster by country. All leads and lags refer to calendar years. For cumulative growth, $h = 1$ is two years and $h = 2$ is three years, including the current observation at $t$.

The coefficient $\beta_h$ is the slope at mean conditioning states, not necessarily the population-average marginal effect in the unbalanced estimation sample. The plotted slope at stress $f$ is $\beta_h + \phi_{F,h} f$; its variance includes $2f\,\mathrm{Cov}(\hat{\beta}_h, \hat{\phi}_{F,h})$. Omitting this covariance is not generally conservative. Country clustering allows within-country dependence but does not eliminate residual regional dependence; a Driscoll–Kraay check is reported in the appendix. Leave-region-out estimates assess sensitivity to regional composition; they are not a substitute for spatially robust inference.

OpenAI Codex assisted with manuscript organization, literature searches, explanatory text, analytical derivations, and implementation and debugging of the estimation, validation and plotting code. The reported statistics and figures are computed from the documented data by that code; AI-generated numerical answers are not used as empirical observations or estimates.

## 4.2 Index persistence

**Table 2:** Conditional index dynamics

| Regressor | Estimate | SE |
|---|---|---|
| Index persistence | 0.864*** | (0.054) |
| Regional association | 0.184*** | (0.048) |
| Banking stress | 0.002 | (0.008) |
| Stress × latent index | -0.011* | (0.006) |
| Inflation | -0.044 | (0.040) |

Country and year effects; 2,090 observations from 135 countries. Stress is standardized within country. The persistence adjustment discussed in the text is an uncapped approximation.

The state equation uses 2090 observations from 135 countries. Estimated persistence is 0.864 (SE 0.054), with a scalar latent-index half-life of 4.7 years conditional on other regressors. Applying the rough Nickell adjustment $\hat{\rho} + (1 + \hat{\rho})/\bar{T}$ gives 0.985, without capping the result. This approximation is not a validated correction for the full interacted, unbalanced specification. An adjusted value at or above one has no finite stationary half-life under Proposition 1. For comparison, $\rho = 0.99$ would imply about 69 years. Equation (3) explains the sensitivity of this transformation near unity.

The sum of the fitted baseline persistence and regional coefficient is 1.049. In the simplified homogeneous endogenous-network model, a sum above one fails the stability condition. The

fitted equation includes fixed effects and stress interactions, so this calculation is a diagnostic, not a complete network estimate. It rules out treating the scalar estimate alone as evidence that the whole regional system is stationary. Neither the index half-life nor its bias adjustment identifies the duration of individual false narratives.

### 4.3 Mean associations and fragility interactions

**Table 3:** Conditional associations and banking-stress interactions

| Outcome | $h$ | $M$ | SE | $M \times F$ | SE | $n$ |
|---|---|---|---|---|---|---|
| Growth | 0 | -0.95*** | (0.35) | -0.14 | (0.17) | 1,735 |
| | 1 | -1.48*** | (0.54) | -0.45** | (0.21) | 1,642 |
| | 2 | -1.52** | (0.73) | -0.94*** | (0.29) | 1,539 |
| | 3 | -1.40 | (0.93) | -1.35*** | (0.36) | 1,434 |
| | 4 | -0.78 | (1.02) | -1.75*** | (0.48) | 1,332 |
| Investment | 0 | -1.44 | (1.07) | 1.06 | (1.30) | 1,628 |
| | 1 | -1.28 | (1.58) | -0.30 | (1.42) | 1,537 |
| | 2 | -0.37 | (2.16) | -0.91 | (1.97) | 1,439 |
| | 3 | 2.26 | (2.59) | -1.44 | (2.40) | 1,338 |
| | 4 | 4.40 | (2.96) | -2.05 | (2.57) | 1,241 |
| Inequality | 0 | -0.09 | (0.14) | 0.04 | (0.05) | 906 |
| | 1 | -0.13 | (0.23) | 0.15 | (0.09) | 877 |
| | 2 | -0.23 | (0.31) | 0.19 | (0.12) | 794 |
| | 3 | -0.49 | (0.33) | 0.13 | (0.17) | 745 |
| | 4 | -0.69* | (0.38) | -0.01 | (0.16) | 673 |
| Exports | 0 | -2.80*** | (0.88) | -0.25 | (0.45) | 1,661 |
| | 1 | -4.15*** | (1.48) | -0.70 | (0.59) | 1,567 |
| | 2 | -3.43* | (1.82) | -1.64** | (0.76) | 1,467 |
| | 3 | -2.90 | (2.46) | -1.69** | (0.80) | 1,363 |
| | 4 | -1.35 | (2.82) | -1.69 | (1.10) | 1,263 |
| NPLs | 0 | 0.28 | (0.22) | 0.23 | (0.20) | 1,663 |
| | 1 | 0.71 | (0.45) | 0.63** | (0.29) | 1,562 |
| | 2 | 1.30** | (0.65) | 1.06*** | (0.40) | 1,462 |
| | 3 | 1.20* | (0.67) | 0.98*** | (0.33) | 1,357 |
| | 4 | 0.93 | (0.67) | 0.83*** | (0.29) | 1,258 |

Growth, investment and exports sum annual growth from $t$ through $t + h$ (percentage points); Gini and NPL outcomes are observed changes from $t - 1$ through $t + h$. Thus $h = 2$ includes three annual observations. Country and year effects; country-clustered SEs. $M$ coefficient is evaluated at mean conditioning states. $^{***}$,$^{**}$,$^{*}$ denote 1, 5 and 10 percent significance. No outcome interpolation.

The three-year growth slope at mean conditioning states is $-1.52$ percentage points (SE 0.73), with a stress interaction of $-0.94$ points (SE 0.29). This regression uses 1,539 observations from 121 countries. At one standard deviation above mean banking stress, their sum is about $-2.47$ points; at one standard deviation below it, their difference is about $-0.58$ points, keeping other conditioning states at their reference values. These are evaluations of the fitted slope per standardized index unit, not predicted growth levels for particular countries. Their uncertainty depends on the joint covariance of the two estimates. The threshold aggregation in Proposition 3 supplies a possible explanation for the sign of this stress gradient, while leaving its scale and causal interpretation to additional measurement and identification. Table 3 reports uncertainty for every outcome and horizon. These are conditional associations with a persistent index, rather than impulse responses to an exogenous innovation. Contemporaneous macroeconomic controls and lagged conditioning states may themselves respond to disinformation or common

shocks, which further prevents a total causal interpretation.

The investment, inequality and NPL estimates are supplementary evidence. A fall in measured inequality is not an increase in inequality harm, an insignificant estimate is not evidence of an exact zero, and average NPL associations cannot establish a stress-feedback mechanism. Interpretation of the financing hypothesis rests on the sign and uncertainty of the growth interaction, its stability across specifications, and the limitations of NPLs as a proxy for financing pressure. At $h = 2$, the no-fill growth slope is $-0.45$ (SE 0.74), while its stress interaction remains negative ($-1.02$, SE 0.27). In the post-2015 sample the interaction is $-0.10$ (SE 0.61), and it is imprecise when Europe and Central Asia are omitted. Thus the baseline evidence does not establish a stable contemporary effect across every sample or conditioning choice.

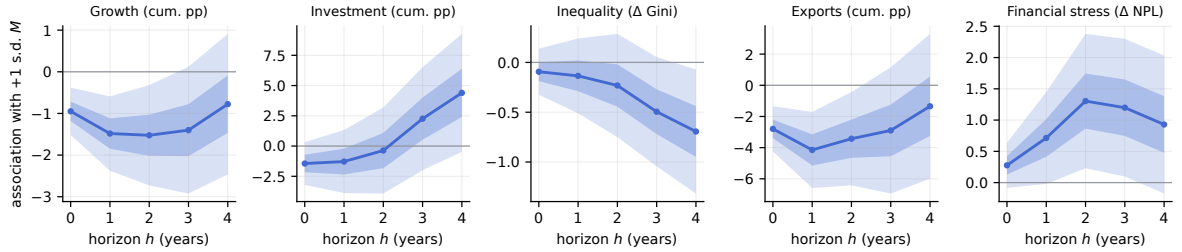



**Figure 2:** Conditional mean slopes at mean conditioning states. Bands are 50 and 90 percent normal-approximation intervals from country-clustered standard errors. Gini and NPL outcomes use observed endpoints.

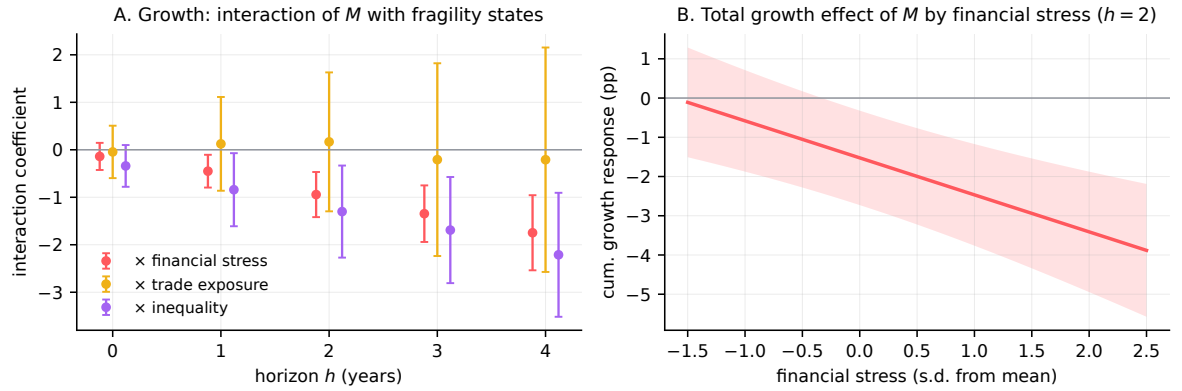



**Figure 3:** Growth interactions and the slope conditional on banking stress. The right panel includes the estimated covariance between the index and its stress interaction. Curves describe an estimated regression surface, not a policy treatment response.

## 4.4 Conditional quantiles and joint uncertainty

For two-year cumulative growth, the first stage fits a two-way fixed-effects mean regression on the index, NPLs, clipped inflation, lagged growth, log GDP per capita and governance. Additive country and year effects are recovered by backfitting and removed from the outcome. The second stage fits pooled quantile regressions on the specified covariates and an intercept, following the location-shift logic of Canay (2011). It is not quantile regression of within-demeaned outcomes on within-demeaned regressors.

The assumption that country and year effects are common location shifts across quantiles is substantive and cannot be justified merely because the interest is in slopes. The procedure estimates a conditional distribution, not the outcome that the same country would experience under a treatment. Its covariate set is simpler than the mean-interaction specification; the results are complementary rather than numerically identical estimands.

The implementation uses a linear-programming quantile solver and 399 country-block bootstrap draws. Each draw resamples complete country histories with replacement, assigns new identifiers to repeated countries, and re-estimates both the mean-effect and quantile stages. Ninety-percent percentile intervals and paired slope contrasts incorporate the first-stage estimation step. They condition on the constructed index; expert-coding uncertainty and residual cross-country dependence are not fully included. The point estimates at the tenth,

**Table 4:** Two-year cumulative growth: conditional quantile slopes

| Quantile | Slope | 90% interval | $n$ |
|---|---|---|---|
| 10th | -0.71 | [-1.68, 0.61] | 2,086 |
| 25th | -0.61 | [-1.58, 0.44] | 2,086 |
| 50th | -0.55 | [-1.49, 0.44] | 2,086 |
| 75th | -0.65 | [-1.58, 0.39] | 2,086 |
| 90th | -0.82 | [-1.81, 0.21] | 2,086 |

399 country-block bootstrap replications, refitting both stages. Intervals condition on the observed index and additive country/year location effects.

**Table 5:** Tests of differences between quantile slopes

| Contrast | Difference | 90% interval | $p$ |
|---|---|---|---|
| 10th minus 50th | -0.17 | [-0.58, 0.48] | 0.575 |
| 10th minus 90th | 0.11 | [-0.44, 0.92] | 0.805 |

Paired country-block draws preserve covariance between quantile estimates; $p$ values use the centered bootstrap null distribution. These are pointwise tests, without multiplicity adjustment.

median and ninetieth quantiles are $-0.71$, $-0.55$ and $-0.82$. Whether the lower-tail slope is distinguishable from the center or upper tail is assessed directly in Table 5. The paired tests do not reject equal slopes (tenth versus median, $p = 0.575$; tenth versus ninetieth, $p = 0.805$). All reported quantile slope intervals include zero. The data therefore do not support a statistically distinct lower-tail association. The location-scale example in Section 2.3 shows why this finding need not contradict amplification conditional on banking stress.

### 4.5 Reverse timing and identification limits

Country and year effects remove additive persistent differences and common year shocks. They do not remove time-varying institutional deterioration, conflict, political instability or reverse causality. Associations with prior outcomes are reported in Appendix B. For negative horizons the code omits lagged GDP growth and outcome-specific lag controls to avoid mechanical overlap with the prior outcome windows. Lagged conditioning states remain in these regressions. Because the index is persistent and common shocks can affect both prior outcomes and current information conditions, these reverse-timing regressions are diagnostics, not experimental pre-trend tests. Omitting the overlapping outcome controls does not establish the absence of confounding.

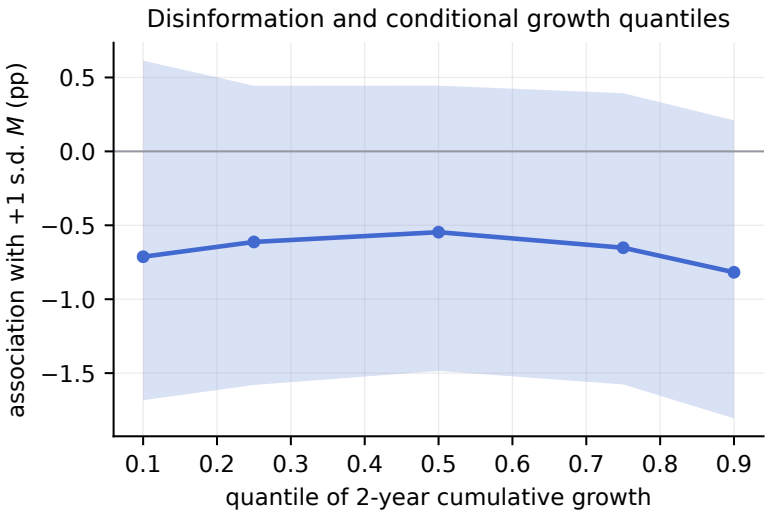



**Figure 4:** Conditional quantile slopes with 90 percent country-block bootstrap intervals. Both estimation stages are refitted in each draw.

A regional-pressure instrument interacted with internet penetration would require an exclusion restriction beyond the fixed effects and controls. Internet exposure may affect growth through other channels, and regional pressure may reflect regional shocks. Exposure measured over 2000–2004 is also not predetermined for observations within that window. Instrument relevance alone would not resolve these concerns. The empirical conclusions therefore rest on conditional associations rather than instrumental-variable identification.

### 4.6 Chronological forecast evaluation

At each annual origin $t = 2014, \ldots, 2023$, the exercise predicts GDP growth in $t + 1$ using covariates from $t - 1$. Training targets end no later than $t - 1$. This conservative timing prevents use of current-year realized growth or a future outcome in model fitting. Each model uses the same observed sample and country intercepts; no future year effect is estimated.

The benchmark contains growth, clipped inflation, log GDP per capita, trade openness, NPLs, private credit and government effectiveness. Missing NPL and credit predictors can be carried forward for at most two years, never backward. The augmented model adds the DSP index. All five component standardizations, the rank transform, and regression scaling are fitted anew using training observations only. Models estimate the conditional mean and the tenth, median and ninetieth quantiles directly. The benchmark and augmented models use identical training and evaluation samples at each origin.

The exercise yields 1,111 held-out country-year predictions. The tenth-quantile loss gain is −3.22 percent and the mean-squared error gain is −0.14 percent. Negative gains denote deterioration. None of the four loss metrics improves at the point estimate; the tenth-quantile gain interval includes zero. These results do not establish operational forecasting value for this index and benchmark. Country-specific intercepts with sparse training histories and omitted common-shock predictors may also limit performance. Intervals resample complete target years, preserving dependence among countries in a given year, but ten target years provide limited precision and do not fully address serial dependence across years. The exercise is pseudo out of sample: source vintages may contain revisions, expert scores may be retrospective, and their

**Table 6:** Pseudo out-of-sample forecast comparison

| Loss | Benchmark | With index | Gain (%) | 90% interval |
|---|---|---|---|---|
| Pinball 0.1 | 1.289 | 1.331 | -3.22 | [-5.44, 1.29] |
| Pinball 0.5 | 1.535 | 1.541 | -0.41 | [-1.57, 0.41] |
| Pinball 0.9 | 0.884 | 0.904 | -2.30 | [-5.04, -0.51] |
| Mean squared error | 23.440 | 23.474 | -0.14 | [-0.53, 0.43] |

Positive gain means lower loss. Identical evaluation samples; target years 2015–2024. Predictor availability lag is one year; training targets end before the forecast origin. Components, index ranks and scaling are fitted within training windows. Intervals resample entire target years (1,999 draws; only ten year blocks). Latest-vintage inputs make this a pseudo rather than real-time evaluation.

actual publication calendars are not reconstructed. A useful association in the panel therefore need not imply an operational early-warning improvement.

## 5 What Can Be Quantified as Risk?

The defensible quantitative outputs are conditional growth profiles, uncertainty intervals, observed banking-stress interactions, and held-out forecast losses. At an observed stress value $f$, the reported profile is $\hat{\beta}_h + \hat{\phi}_{F,h} f$, with other conditioning states at their reference values. It should be evaluated within the empirical support and with its covariance-based uncertainty. A change in this profile is not an estimated intervention effect.

The horizon convention also matters for economic interpretation. The growth outcome at $h = 2$ adds three annual growth rates. A slope of $-1.52$ percentage points is therefore a difference in that sum, holding regressors fixed; it is not an annual growth penalty to be applied three more times. Compounded output growth instead depends on the product of annual gross growth factors, and recovering it requires a path of annual rates. Moreover, the horizon-specific regressions can use different samples, so differences between their cumulative slopes need not recover a common annual profile.

Projecting a sustained change in the index into a structural growth scenario would require explicit dynamic equations, shock identification, and validation of the mapping from interventions to parameters. Those ingredients are not supplied by the current panel.

Combining growth, inequality and banking outcomes into a single burden score requires economically meaningful units and weights. Normalizing a channel by its estimated two-year response is unstable when that response is near zero and uncertain. Allowing signed outcomes can also reverse a ranking based on banking stability alone. A bounded mathematical score does not establish welfare meaning, comparability, or reliable policy rankings. The appendix retains the elementary boundedness result solely to document the mathematical scope of this construction. No GDP-loss totals, burden reductions, or empirical trap thresholds are inferred from it.

## 6 Policy Interpretation and Research Requirements

The results motivate scrutiny of the information environment alongside financial fragility, subject to the predictive validation in Table 6. They do not establish that collecting the proposed index improves decisions, that a financial rumor persists for the aggregate index's

half-life, or that a measured country is near a crisis threshold.

Rapid official communication, message authentication and restrictions on content propagation are distinct candidate interventions. Their effects may span the model intercept, persistence, innovation variance, public trust and the reach of accurate information. Ranking them requires estimates of these responses, implementation costs and possible adverse effects. The conditional elasticity result in Proposition 5 does not provide those estimates. Equation (11) makes the missing comparison explicit.

A stronger policy study would link independently identified financial falsehoods to audience exposure, bank funding or lending outcomes and credible correction timings. It would then estimate the response to an intervention using a design with a defensible counterfactual, rather than relabeling a country-level association. A financial-content validation sample would also test whether the broad DSP proxy captures the mechanism proposed here. Out-of-time and real-time validation against richer financial-condition, uncertainty and geopolitical-risk benchmarks would be needed before proposing an operational monitoring instrument. These are requirements for stronger claims, not results claimed by the present paper.

# 7 Conclusion

A broad disinformation index can be examined as a correlate of macroeconomic risk without equating political-information scores with financial-message contamination. The panel estimates a three-year cumulative-growth slope of $-1.52$ percentage points at mean conditioning states and a banking-stress interaction of $-0.94$. Quantile inference and chronological prediction evaluate the scope of the evidence, including the possibility that in-sample associations do not improve held-out forecasts.

The illustrative financing model supplies an increasing-differences result under an explicitly assumed lending rule. Feedback can generate multiple steady states for some parameters, but the empirical exercise does not identify a trap threshold. Persistence comparative statics rank parameter changes only under sign and effectiveness restrictions; they do not identify policy cost-effectiveness. The remaining contribution is a reproducible conditional analysis, with measurement, identification and forecasting limitations made explicit. Financial-specific data and credible intervention variation are the next steps needed to support causal policy conclusions.

Selected algebraic results have companion Lean 4 formalizations against mathlib (de Moura and Ullrich 2021; The mathlib Community 2020); their scope is specified below.

# A Proofs and Formalization Scope

## A.1 Scalar state dynamics

The fixed-point equation $x = c + \rho x$ gives $\mu = c/(1-\rho)$. Subtracting it from the recursion yields $x_{t+1} - \mu = \rho(x_t - \mu)$, which proves the closed form and convergence for $|\rho| < 1$. Logistic bounds and strict monotonicity follow from $e^{-x} > 0$. With independent innovations, the stationary representation is $x_t = \mu + \sum_{j \geq 0} \rho^j u_{t-j}$ and the variance follows by a geometric sum. Solving $\rho^h = 1/2$ gives the latent half-life formula. These statements condition on absent or fixed external inputs; they are not a stationarity proof for the fitted regional panel.

For fixed nonnegative row-stochastic $W$, the vector of ones is an eigenvector with eigenvalue one. Hence $\rho + \delta$ is an eigenvalue of $A = \rho I + \delta W$. The induced infinity norm is $\rho + \delta$, so the spectral radius is exactly that value for nonnegative $\rho, \delta$. A time-varying network or country-dependent interaction coefficients require a separate stability analysis.

For equation (2), apply the mean-value theorem between $\mu$ and $\mu + \rho^t d_0$. The derivative bound $\sigma' \leq 1/4$ gives the stated inequality; a first-order expansion at $\mu$ gives the local approximation. Differentiating $h(\rho)$ yields equation (3), and $-\log(\rho)/(1-\rho) \to 1$ gives its limiting expression. These steps explain both the bounded scale of index changes and the sensitivity of half-life transformations without assuming that boundedness guarantees stationarity of the latent process.

## A.2 Signal noise and the lending assumption

Independence gives $\mathbb{E}[b\eta] = 0$ and $\mathrm{Var}(b\eta d) = d^2\mathbb{E}[b\eta^2] = pd^2$. Adding independent $\varepsilon$ proves Lemma 1. The posterior distribution of $V$ has not been specified; the lemma does not compute its variance. Under the assumed credit rule (4),

$$I^* - \min\{I^*, W_0 + L(p)\} = \max\{\lambda\gamma d^2 p + I^* - W_0 - \lambda\bar{V} + \lambda\gamma\sigma_\varepsilon^2, 0\}.$$

Collecting the financing-pressure terms gives (5). This is a conditional derivation from a stated reduced-form lending rule, not a Bayesian microfoundation.

## A.3 Increasing differences

Let $g(z) = \max\{z, 0\}$, $s = ap_1 + bF_1 - \kappa_0$, $u = a(p_2 - p_1) \geq 0$ and $v = b(F_2 - F_1) \geq 0$. The increment $g(s+u) - g(s)$ is nondecreasing in $s$, which implies $g(s+u) - g(s) \leq g(s+v+u) - g(s+v)$. This proves increasing differences. When $ap_2 + bF - \kappa_0 \leq 0$, both terms of the contamination increment are zero. No quantitative claim about mean or quantile growth effects follows without further assumptions on aggregation and other shocks.

### A.4 Aggregation across financing thresholds

To prove Proposition 3, write $z = ap + bF$ and

$$A(z) = \mathbb{E}[(z - K)_+] = \int_{-\infty}^{z} (z - k) f_K(k)\, dk.$$

The integrand is integrable under the finite-moment assumption. Its difference quotient with respect to $z$ is bounded in absolute value by one and converges to $\mathbf{1}\{K < z\}$ except at $K = z$, which has zero probability. Dominated convergence gives $A'(z) = C_K(z)$; continuity of the density then gives $A''(z) = f_K(z)$. The chain rule establishes $\overline{H}_p = aA'(z)$ and $\overline{H}_{pF} = abA''(z)$. Applying it again to $-\chi A(ap(m) + bF)$ proves equation (8). If the threshold distribution has atoms, the smooth derivative expression can fail at those atoms, although the individual increasing-differences inequality still survives averaging over a fixed distribution.

The condition that the distribution is fixed is consequential. For a conditional distribution $C_K(k \mid p, F)$, changes in $p$ or $F$ alter both individual shortfalls and population composition. Differentiating the expectation then introduces distributional terms with unrestricted signs. The proposition therefore identifies a particular constraint channel rather than a universal restriction on observed country-level growth.

For equation (9), positive $s$ preserves the ordering of the realizations of $\epsilon$. Quantile equivariance gives $Q_\tau = \ell + sq_\tau$, so $\partial_m Q_\tau - \partial_m Q_\nu = s_m(q_\tau - q_\nu)$. If $s_m = 0$, all such contrasts vanish while $\ell_{mF}$ can remain negative. This provides an explicit example in which fragility amplification and equal quantile slopes hold simultaneously.

### A.5 Feedback and multiplicity

The map in (10) is globally Lipschitz with constant at most $\rho + \varphi L_g$. When this bound is below one, Banach's fixed-point theorem on the complete space $\mathbb{R}$ gives existence, uniqueness and global convergence. In the piecewise-linear example, solving on each of $x \leq 0$, $0 \leq x \leq 1$ and $x \geq 1$ gives respectively $-3/2$, $1/2$ and $5/2$. Slopes at those roots are $1/2$, $5/2$ and $1/2$. The example proves that multiplicity is possible, not that it begins when an upper bound on a derivative reaches one.

### A.6 Elasticities and the optional burden bound

For $c > 0$ and $0 < \rho < 1$,

$$\frac{\partial \mu}{\partial c} = \frac{1}{1 - \rho}, \qquad \frac{\partial \mu}{\partial \rho} = \frac{c}{(1 - \rho)^2}, \qquad \frac{\partial^2 \mu}{\partial \rho^2} = \frac{2c}{(1 - \rho)^3}.$$

Multiplying the first derivatives by $c/\mu$ and $\rho/\mu$ establishes Proposition 5. The same algebraic elasticity identity extends to $c \neq 0$, but negative $c$ reverses the desired direction of a reduction in persistence. At $c = 0$, division by $\mu$ is undefined. Applying the chain rule to expenditure gives (11).

The marginal comparison can also be distinguished from a finite parameter change. Let $0 < e < 1$ denote an equal proportional reduction, holding the other parameter fixed. The

proportional declines in the positive latent steady state are

$$\frac{\mu - \mu_c(e)}{\mu} = e, \qquad \frac{\mu - \mu_\rho(e)}{\mu} = \frac{e\rho}{1 - \rho + e\rho}, \tag{14}$$

where $\mu_c(e) = (1-e)c/(1-\rho)$ and $\mu_\rho(e) = c/\{1-(1-e)\rho\}$. Since the denominators are positive, the persistence reduction is larger precisely when $\rho > 1/(2-e)$. As $e \downarrow 0$, this threshold tends to $1/2$, recovering the infinitesimal comparison. For positive finite $e$ the threshold is higher. Thus the local elasticity ranking is not a formula for arbitrarily large interventions. Applying the increasing logistic function preserves the ordering of the two deterministic endpoint changes, but not their proportional magnitudes or their welfare interpretation.

For completeness, if $M_h \in [0,1]$, $|Y_h| \leq \bar{Y}$ and $\kappa > 0$, then $D_H = \sum_{h=0}^{H}(1+\kappa)^{-h} M_h Y_h$ satisfies $|D_H| \leq \bar{Y}(1+\kappa)/\kappa$ by the triangle inequality and the geometric-series bound. Signed outcomes permit negative scores. This bound is not a welfare result and does not validate that normalization.

### A.7 What the Lean files establish

The repository includes proofs of logistic bounds and monotonicity; scalar deterministic dynamics and a geometric variance identity; increasing differences of the max-linear harm function; uniqueness given a contraction bound and an explicit three-root example; an absolute burden bound; and algebraic derivative, elasticity and midpoint-convexity results. These are selected mathematical statements, not a machine verification of all economic interpretations. The aggregation, location-scale and finite-change derivations above are analytical extensions; they are not included in the companion Lean formalizations. In particular, the files do not establish construct validity, posterior-risk microfoundations, empirical identification, forecast performance, intervention effectiveness, stochastic ergodicity of the full panel, or a data-calibrated bifurcation threshold. The formalization mapping and build instructions are recorded in the repository.

## B Additional Empirical Evidence

**Table 7:** Associations with prior outcomes

| Outcome | $h$ | Coefficient | SE | $n$ |
|---|---|---|---|---|
| Growth | -2 | -1.18** | (0.54) | 1,734 |
| Growth | -1 | -0.73** | (0.33) | 1,735 |
| Investment | -2 | -2.08 | (1.96) | 1,630 |
| Investment | -1 | -2.13* | (1.25) | 1,638 |
| Inequality | -2 | 0.14 | (0.21) | 987 |
| Inequality | -1 | 0.12 | (0.13) | 958 |
| Exports | -2 | -2.63** | (1.12) | 1,662 |
| Exports | -1 | -2.76*** | (0.92) | 1,671 |
| NPLs | -2 | 0.24 | (0.59) | 1,488 |
| NPLs | -1 | 0.17 | (0.28) | 1,594 |

Past growth outcomes sum years $t+h$ through $t-1$; past level changes run from $t+h-1$ to $t-1$. Lagged outcomes are omitted to avoid mechanical overlap. These are reverse-timing diagnostics, not clean falsification tests for a persistent index.

**Table 8:** Growth associations under alternative specifications ($h = 2$)

| Specification | $M$ | SE | $M \times F$ | SE | $n$ |
|---|---|---|---|---|---|
| dk | -1.52*** | (0.39) | -0.94** | (0.39) | 1,539 |
| domestic_only | -1.56** | (0.70) | -0.88*** | (0.27) | 1,539 |
| end2022 | -0.78 | (0.84) | -0.72** | (0.33) | 1,232 |
| equal_weights | -1.43* | (0.75) | -0.99*** | (0.30) | 1,539 |
| index_history | -2.44*** | (0.77) | -0.96*** | (0.29) | 1,539 |
| no_fill | -0.45 | (0.74) | -1.02*** | (0.27) | 1,041 |
| Omit East Asia & Pacific | -1.57** | (0.78) | -1.05*** | (0.31) | 1,361 |
| Omit Europe & Central Asia | -1.69 | (1.20) | -0.90 | (0.88) | 879 |
| Omit Latin America & Caribbean | -2.28*** | (0.79) | -0.93*** | (0.31) | 1,297 |
| Omit Middle East & North Africa | -1.68** | (0.72) | -1.00*** | (0.29) | 1,447 |
| Omit North America | -1.63** | (0.75) | -0.92*** | (0.29) | 1,507 |
| Omit South Asia | -1.27* | (0.72) | -0.91*** | (0.30) | 1,484 |
| Omit Sub-Saharan Africa | -0.67 | (0.77) | -0.85*** | (0.26) | 1,259 |
| post2015 | -1.51* | (0.81) | -0.10 | (0.61) | 839 |
| raw_composite | -1.45* | (0.83) | -0.94*** | (0.34) | 1,539 |

No-fill uses observed conditioning states only. Raw-composite uses the unranked weighted index; each index is standardized in its specification. Alternative weights and samples can change results; stability is not presumed.

Alternative-weight specifications reconstruct the index from the published components. The raw-composite check removes the empirical-CDF transformation. Sample restrictions, no-fill conditioning, a lagged-index control, Driscoll–Kraay inference and leave-region-out estimates are explicitly recomputed. Their signs, uncertainty and sample sizes should be compared with the baseline rather than summarized as uniformly robust. The Driscoll–Kraay calculation applies a Bartlett kernel with bandwidth two to year-aggregated within-regression scores; inference uses year-count minus one degrees of freedom. The short time dimension limits this approximation. A lagged-index specification asks whether the current index adds information conditional on its recent history, which is a different estimand from the baseline level association.

The Monte Carlo exercise simulates the repository's own stylized linear-interaction data-generating process in 200 replications. Its output is retained as a numerical estimator diagnostic. It does not test the DSP measurement process, time-varying confounding, the forecasting exercise or the assumed financing mechanism. Bias correction reduces but need not eliminate dynamic-panel bias.

## C Synthesis of the Analytical and Empirical Results

The proofs give a coherent interpretation of the evidence when the information environment, financing constraints and growth distribution are kept conceptually distinct. Persistence describes how an aggregate latent index evolves. The constraint model describes how signal noise can restrict investment, and its aggregation shows why the concentration of borrowers near a financing boundary matters for amplification. Neither persistence nor amplification alone determines the relative slopes of growth quantiles.

This distinction helps reconcile the numerical findings. The negative baseline growth slope and banking-stress interaction are compatible with a financing channel, subject to the proxy mapping and the sensitivity to sample and conditioning choices. The quantile contrasts do not establish an additional lower-tail concentration of the association. The forecast comparison asks a further question—whether the index contributes useful information beyond available macro-financial predictors—and finds no improvement in the evaluated design. These results address different properties of the data, so agreement on every dimension is not required for a coherent interpretation.

The policy implications remain conditional on the same structure. Persistence can magnify a positive latent steady state, but the sign of the intercept, the size of the parameter change and the cost of changing it determine any intervention ranking. A mathematical possibility of multiple steady states does not locate an empirical crisis threshold. The combined analysis thus provides a basis for testing financial-information exposure and financing constraints more directly, while keeping the measured associations separate from causal welfare calculations.

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